EE 2305 – Introduction to C Programming
Hardware Project 03
Digital VU Meter

Project Features: Analog Input and Digital Output.

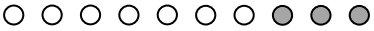
Program an Arduino board to accept an analog input voltage and display the magnitude of the voltage using a 10 LED display.



Input voltage = 0% of maximum (all LEDS off)



Input voltage = 30% of maximum

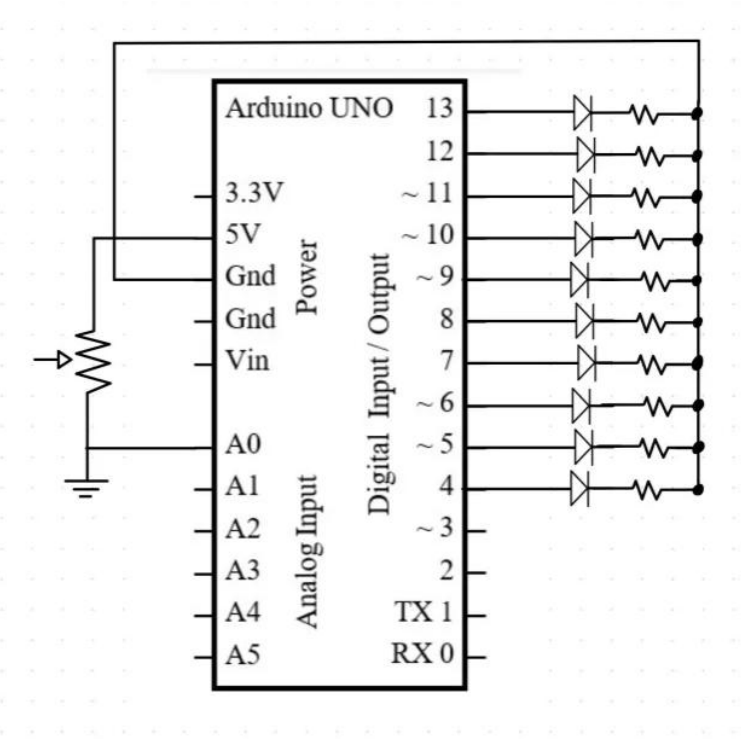


Input voltage = 70% of maximum

Document your program and include the following sections. Provide a brief description of the system and how you are designing it to operate.

A. Hardware Diagram:

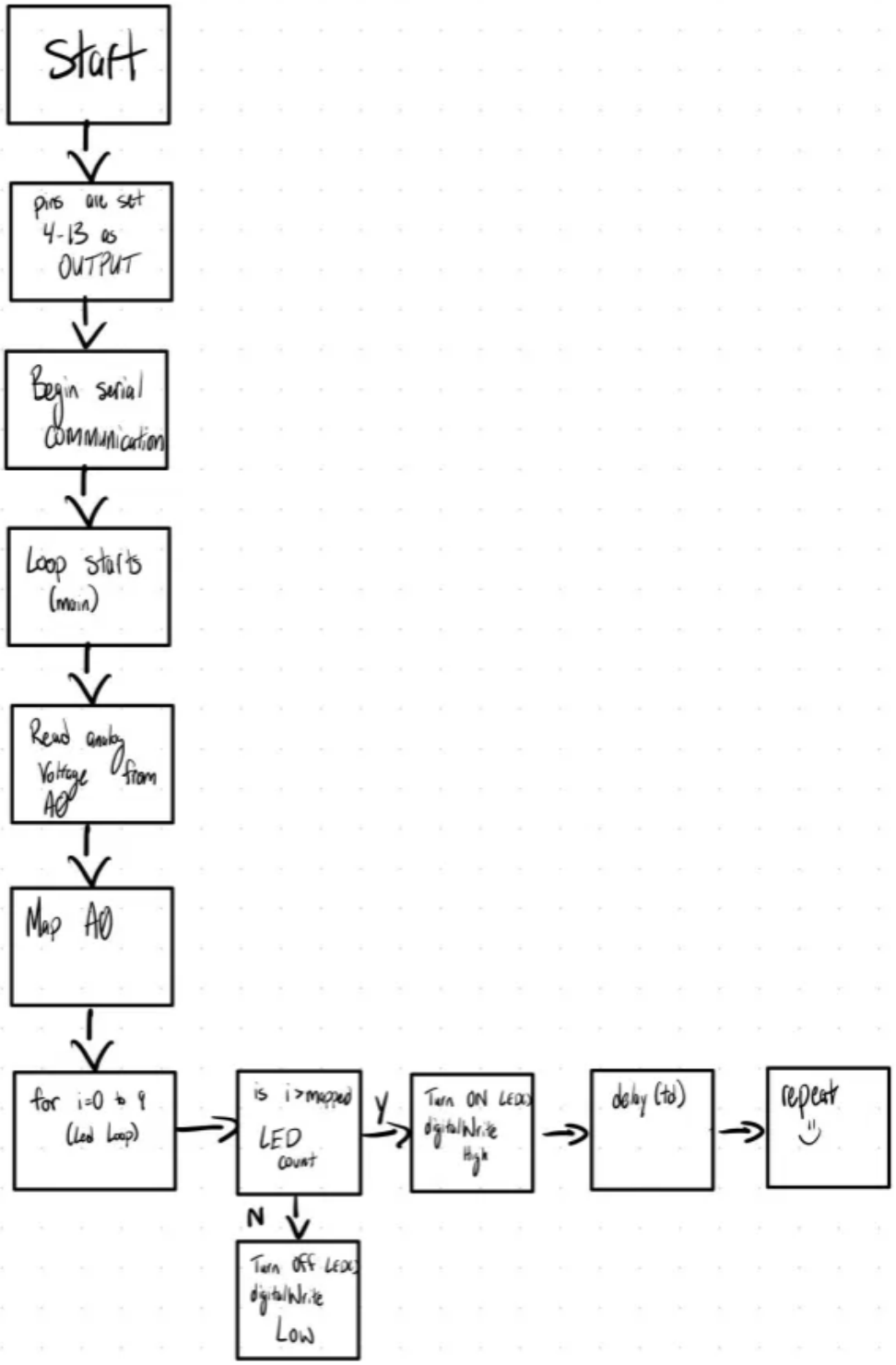
Provide a hardware diagram of the components.



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B. Program Flowchart:

Draw a flowchart of the program.



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C. Arduino Source Code

Insert the Arduino Source Code into the document.

```
// Digital VU Meter - ECE 2305 Hardware Project 03

const int td = 100;
int analogVoltage = 0;

void setup() {
  for (int pin = 4; pin <= 13; pin++) {
    pinMode(pin, OUTPUT);
  }
  Serial.begin(9600);
}

void loop() {
  analogVoltage = analogRead(A0);
  Serial.println(analogVoltage);

  int ledCount = map(analogVoltage, 0, 1023, 0, 10); // Map analog input to 0-10
LEDs

  for (int i = 0; i < 10; i++) {
    if (i < ledCount) {
      digitalWrite(4 + i, HIGH); // Turn on LEDs up to ledCount
    } else {
      digitalWrite(4 + i, LOW);  // Turn off the rest
    }
  }

  delay(td);
}
```

D. Demonstration Video

Record and upload a video demonstrating the operation of the circuit.
DONE
Save the document as a *PDF* file and submit the *PDF* document to *Blackboard*.